



X-ray Doses in Relation to Body Mass, Indication, and Substrate During Pediatric Electrophysiological Procedures on the Heart

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Abstract

The main goal of this study is to determine typical values of dose area product (DAP) and difference in the effective dose (ED) for pediatric electrophysiological procedures on the heart in relation to patient body mass. This paper also shows DAP and ED in relation to the indication, the arrhythmia substrate determined during the procedure, and in relation to the reason for using radiation. Organ doses are described as well. The subjects were children who have had an electrophysiological study done with a 3D mapping system and X-rays in two healthcare institutions. Children with congenital heart defects were excluded. There were 347 children included. Significant difference was noted between mass groups, while heavier children had higher values of DAP and ED. Median DAP in different mass groups was between 4.00 (IQR 1.00–14.00) to 26.33 (IQR 8.77–140.84) cGycm². ED median was between 23.18 (IQR 5.21–67.70) to 60.96 (IQR 20.64–394.04) μSv. The highest DAP and ED in relation to indication were noted for premature ventricular contractions and ventricular tachycardia—27.65 (IQR 12.91–75.0) cGycm² and 100.73 (IQR 53.31–258.10) μSv, respectively. In arrhythmia substrate groups, results were similar, and the highest doses were in ventricular substrates with DAP 29.62 (IQR 13.81–76.0) cGycm² and ED 103.15 (IQR 60.78–266.99) μSv. Pediatric electrophysiology can be done with very low doses of X-rays when using 3D mapping systems compared to X-rays-based electrophysiology, or when compared to pediatric interventional cardiology or adult electrophysiology.

Keywords Electrophysiology · Heart · Children · X-rays · Radiation dose · Typical doses

Abbreviations

3D	Three-dimensional
AT	Atrial tachycardia
AP	Accessory pathways
CF	Conversion factor
DAP	Dose area product
ED	Effective dose
Gy	Gray
IQR	Interquartile range
MS	Mapping system
Sv	Sievert

Introduction

In electrophysiology (EP), the gold standard for displaying the position of a catheter in the patient's body is using X-rays. Determining the dose of radiation a patient receives, especially the impact of that radiation on individual tissues and organs, is very important.

The typical value of a particular dosimetric quantity shows the usual clinical practice, that is, the way a particular procedure is carried out using a particular modality. In the case when diagnostic reference levels are available, typical values are compared with them and, if necessary, corrective measures are taken in order to optimize the procedure. In the case of adult patients, typical values are calculated for the so-called standard patient (70 ± 20 kg) [1], while in the case of pediatric patients, it is recommended to show them for the following groups according to the patient's weight: 0–< 5 kg, 5–< 15 kg, 15–< 30 kg, 30–< 50 kg, and 50–< 80 kg [2]. Knowing the value of a particular dosimetric quantity also allows the estimation of the dose of a

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particular organ or the effective dose (ED) using the appropriate conversion factors given in the literature [3]. It should be noted that ED in medicine is used only for the purposes of comparing different procedures, individual modalities, and procedures in different health institutions and/or countries, as well as for comparing different modalities for the implementation of the same procedure [4].

It is important to emphasize that in EP procedures, a smaller child brings more challenges during work. The puncture of the blood vessel itself is more demanding, catheter placement and mapping of smaller structures are more challenging, and the chance of complications is greater. It is also a question of the operator's trust in the accuracy of the 3D mapping system when precision is extremely important for smaller structures. Therefore, in the case of EP procedures in smaller children, one could expect more frequent use of X-rays, as well as the need for better image quality and, consequently, greater radiation exposure.

To the best of our knowledge, although 3D mapping systems for displaying catheter position have been in use for years, there are no studies that provide typical values of dose area product (DAP), estimated organ doses, and ED for pediatric EP procedures for cases where X-ray radiation is also used, for specific categories of patients based on mass. Certain articles showed that the fluoroscopy time, as well as the typical value of a certain dosimetric quantity, was significantly reduced after the introduction of a 3D mapping system, but without distinguishing patients according to mass [5–16]. To the best of our knowledge, there are no studies in which the estimated organ doses and ED for these procedures are given. Several studies described that in some EP procedures performed using a 3D navigation system, it is possible to completely avoid X-rays [17–19].

These data are missing when determining guidelines in the treatment of arrhythmias, but also when providing information to patients or their caregivers. The aim of this study was to determine the typical values of DAP and ED for the aforementioned groups of pediatric patients which are based on the patient's weight. Furthermore, DAP and ED were compared for different indications, and the arrhythmia substrate was determined during the procedure and for the reason of using X-rays. Organ doses have also been estimated.

Methods

The research is a multicentric cross-sectional study. Subjects were children under the age of 18 years of age who underwent a pediatric EP study with or without ablation during the period from January 1, 2016, until December 31, 2021, at Gottsegen Cardiovascular Center, Budapest, Hungary, and from January 1, 2018, until December 31, 2021, at University hospital center Sestre milosrdnice, Zagreb, Croatia.

Children with all types of arrhythmias were included. Procedures conducted without the use of X-rays were excluded since the aim of this study was to investigate the DAP and ED when it is used. Children with congenital heart defects were also excluded, as they represent a much smaller group that has its own specifics about the need to use X-rays [20].

Most of the procedures were performed under propofol sedation, except for children in Zagreb over the age of 12, who were performed under local anesthesia.

All procedures were done with the use of 3D mapping systems. Abbott's Ensite NavX system and Johnsons & Johnsons Carto system were used. The X-ray unit used at the Gottsegen Cardiovascular Center was General Electrics INNOVA 5 and in Zagreb Siemens CIOS Alpha. All cases were done with pediatric protocol on X-ray units and the automatic exposure control was used.

All the procedures were performed with the intention of performing a safe and effective ablation while maintaining the patient exposure as low as possible without compromising the outcome of the procedure. The left-sided accessory pathways were targeted via an open foramen ovale or a transseptal approach using fluoroscopy for puncture. We did not routinely use additional imaging modalities. Only in the case of difficult transseptal punctures, intracardiac echocardiography or transoesophageal ultrasound was used. Puncture of a vein or artery, introduction of wires and placement of introducers were done without the systematic use of X-rays, except in case of a difficult puncture. Short introducers were usually used, and a long introducer was used for transseptal punctures and when necessary to improve catheter stability.

For each patient, the following data were collected from the medical records: demographic and clinical information (weight, sex, presence of structural heart disease, indication for the EP study) and the findings of the electrophysiological study (arrhythmia substrate, reason for fluoroscopy, DAP value).

ED and organ doses were calculated as described in Buytaert et al. [21] with conversion factors (CF) from the same article. The patients were classified into groups according to their weight: (1) 15–< 30 kg, (2) 30–< 50 kg, (3) 50–< 80 kg, and (4) $\geq$ 80 kg. Since the number of patients in pediatric electrophysiology with a mass less than 15 kg was too small for a separate group, typical values are also shown using a curve as it was previously done [22].

Indications for the electrophysiological procedure were classified into the following groups: palpitations, recorded supraventricular tachycardia (SVT), asymptomatic preexcitation, Wolff–Parkinson–White (WPW) syndrome and premature ventricular contractions (PVC)/ventricular tachycardia (VT).

The substrate determined by electrophysiological study and possibly confirmed by ablation was classified into the following groups: right-sided accessory pathways (AP),

left-sided AP, atrioventricular nodal reentry tachycardia (AVNRT), atrial tachycardia (AT), and ventricular substrate.

Reasons for fluoroscopy were classified into categories: (1) transseptal puncture, (2) positioning of the wire, introducer or catheter, (3) mapping with catheter, and (4) angiography (coronary sinus angio, coronary angiography, etc.).

Statistical Analysis

Results are based on two-sided tests. Normality of continuous data distribution has been checked with Kolmogorov–Smirnov test and according to the results, appropriate non-parametric test has been used in followed analyses. Kruskal–Wallis test was used for radiation doses comparison between age groups with additional Dunn post hoc tests for any significant findings. For Box and Whiskers plot and scatter diagram, additional logarithmic transformation of radiation dose has been previously applied. All p values below 0.05 have been considered significant. Due to non-Gaussian distribution and several outliers, logarithmic transformation had been applied to radiation doses for graphical presentation (Box and Whiskers plot).

Data were analyzed with MedCalc® Statistical Software version 20.210 (MedCalc Software Ltd, Ostend, Belgium; <https://www.medcalc.org>; 2022).

Results

Patient Demographics

The research included a total of 347 children who underwent an electrophysiological study with the use of the 3D mapping system and X-rays in the centers and period.

The research included 203 (58.5%) male and 144 (41.5%) female children. The average age was 13.03 years.

In the 15–<30 kg group, there were 41 patients; in the second group (30–<50 kg), there were 96; and in the third (50–<80 kg), there were 164 and in the fourth (≥ 80 kg) 38. There were 8 children with mass below 15 kg which is too small a number for statistical analysis.

The most common indication for the procedure was SVT in 148 (42.7%) subjects, followed by preexcitation 74 (21.3%), WPW syndrome 67 (19.3%), palpitations 30 (8.6%), and PVC/VT 28 (8.1%).

The most common arrhythmologic substrate determined by electrophysiological study was left-sided accessory pathways in 200 (57.8%) patients. They were followed in frequency by the right-sided accessory pathways in 70 (20.2%), AVNRT in 30 (8.7%), ventricular substrate in 27 (7.8%), and atrial tachycardia in 19 (5.5%) patients.

DAP and ED

The highest median DAP value was in the mass group ≥ 80 kg 26.33 (IQR 8.77–140.84) cGycm². Values for all groups are shown in Table 1 with minimum, maximum, and percentile values. The statistical difference was significant for the groups of higher mass as shown in Table 2.

ED values were larger in higher mass groups. These values are also shown in Table 1. There was a significant statistical difference when comparing the 30–<50 kg group to larger ones (Table 2).

Organ doses are shown in Table 1 with statistical differences in Table 2.

The dependence of DAP value on the patient's mass with the median and third quartile presented as a curve is shown in Fig. 1; children with mass <15 kg were also included.

DAP and ED According to Indication for Electrophysiology

The DAP median observed according to the indication for the electrophysiology procedure was highest in the PVC/VT group with a value of 27.65 (IQR 12.91–75.0) cGycm². This and other values of DAP in relation to the indication are shown in Table 3.

There was a statistically significant difference in DAP between different indications ($p < 0.001$), due to the difference between SVT and preexcitation ($p = 0.009$), SVT and PVC/VT ($p < 0.001$), preexcitation and WPW ($p = 0.001$), preexcitation and palpitation ($p = 0.021$), preexcitation and PVC/VT ($p < 0.001$), WPW and PVC/VT ($p = 0.021$), and palpitation and PVC/VT ($p = 0.003$).

The ED median, as shown in Table 3, was also the highest for PVC/VT with a value of 100.73 (IQR 53.31–258.10) μ Sv. Statistically significant differences were noted for ED in between indications ($p < 0.001$). The difference was noted between SVT and preexcitation ($p = 0.009$), SVT and PVC/VT ($p = 0.001$), preexcitation and WPW ($p = 0.003$), preexcitation and PVC/VT ($p < 0.001$), WPW and PVC/VT ($p = 0.042$), and palpitation and PVC/VT ($p = 0.003$).

DAP and ED According to Arrhythmia Substrate

DAP values divided into categories by arrhythmia substrate were the highest in the ventricular group, with the median value of 29.62 (IQR 13.81–76.0) cGycm². The median values of DAP according to arrhythmia substrates with minimum, maximum, and percentiles are shown in Table 4. There was only one patient in whom the EP study was performed but without induction of tachycardia or signs of AP, so substrate has not been determined.

There was a significant statistical difference ($p < 0.001$) in the value of DAP between different arrhythmia

Table 1 Dose area product, effective dose, and organ doses in relation to mass groups

Weight groups	N	Min	Max	Percentiles			P
				25th	50th (Median)	75th	
<i>Dose area product (cGycm²)</i>							
15-< 30 kg	41	0.01	396.69	1.00	4.00	14.00	<0.001
30-< 50 kg	96	0.01	186.75	1.00	4.99	12.57	
50-< 80 kg	164	0.01	553.00	4.00	12.50	36.72	
> =80 kg	38	0.10	642.49	8.77	26.33	140.84	
<i>Effective dose (μSv)</i>							
15-< 30 kg	41	0.07	1700.90	7.22	27.49	119.408	0.010
30-< 50 kg	96	0.05	1133.13	5.21	23.18	67.70	
50-< 80 kg	164	0.03	1713.97	13.86	43.26	123.12	
> =80 kg	38	0.28	1767.88	20.64	60.96	394.04	
<i>Heart dose (μGy)</i>							
15-< 30 kg	41	0.24	6359.31	25.14	95.65	415.55	0.004
30-< 50 kg	96	0.14	3464.76	15.92	70.89	207.00	
50-< 80 kg	164	0.10	5347.03	43.23	134.96	384.09	
> =80 kg	38	1.00	6304.69	73.59	217.40	1405.25	
<i>Stomach dose (μGy)</i>							
15-< 30 kg	41	0.04	536.92	3.90	14.83	64.44	0.022
30-< 50 kg	96	0.02	536.08	2.46	10.97	32.03	
50-< 80 kg	164	0.01	690.84	5.59	17.44	49.62	
> =80 kg	38	0.12	767.53	8.96	26.47	171.08	
<i>Colon dose (μGy)</i>							
15-< 30 kg	41	0.00	26.66	0.32	1.21	5.25	0.012
30-< 50 kg	96	0.00	37.35	0.17	0.76	2.23	
50-< 80 kg	164	0.00	41.02	0.33	1.04	2.95	
> =80 kg	38	0.01	53.20	0.62	1.83	11.86	
<i>Liver dose (μGy)</i>							
15-< 30 kg	41	0.05	1544.08	5.45	20.72	90.02	0.014
30-< 50 kg	96	0.05	1093.04	5.02	22.36	65.30	
50-< 80 kg	164	0.03	1500.65	12.13	37.88	107.80	
> =80 kg	38	0.25	1600.01	18.68	55.17	356.62	
<i>Lungs dose (μGy)</i>							
15-< 30 kg	41	0.27	3166.98	27.91	106.22	461.5	0.011
30-< 50 kg	96	0.17	4235.93	19.46	86.66	253.07	
50-< 80 kg	164	0.12	6229.87	50.37	157.24	447.50	
> =80 kg	38	1.04	6563.76	76.61	226.33	1462.99	
<i>Urinary bladder dose (μGy)</i>							
15-< 30 kg	41	0.00	2.72	0.04	0.15	0.66	<0,001
30-< 50 kg	96	0.00	3.30	0.02	0.07	0.20	
50-< 80 kg	164	0.00	2.46	0.02	0.06	0.18	
> =80 kg	38	0.00	4.15	0.05	0.14	0.92	
<i>Thyroid dose (μGy)</i>							
15-< 30 kg	41	0.01	74.59	1.10	4.19	17.25	0.015
30-< 50 kg	96	0.01	119.19	0.55	2.44	7.12	
50-< 80 kg	164	0.00	138.66	1.12	3.50	9.96	
> =80 kg	38	0.03	165.80	1.94	5.72	36.96	

substrates. The difference was noted between AVNRT and left-sided AP ($p=0.007$), AVNRT and AT ($p=0.042$), AVNRT and ventricular substrate ($p<0.001$), right AP and

ventricular substrate ($p<0.001$), left AP and ventricular substrate ($p<0.001$), and AT and ventricular substrate ($p=0.016$).

Table 1 (continued)

Weight groups	N	Min	Max	Percentiles			P
				25th	50th (Median)	75th	
Active bone marrow dose (μGy)							
15<30 kg	41	0.04	356.39	3.62	13.78	59.86	<0,001
30<50 kg	96	0.03	765.68	3.58	15.67	45.74	
50<80 kg	164	0.03	1743.51	14.10	44.01	125.24	
>=80 kg	38	0.29	1821.77	21.26	62.82	406.05	
Breasts dose (μGy)							
15<30 kg	41	0.10	675.01	9.98	37.96	164.90	0.125
30<50 kg	96	0.07	1753.25	8.05	35.87	104.75	
50<80 kg	164	0.04	2225.95	18.00	56.18	159.90	
>=80 kg	38	0.30	1882.57	21.97	64.92	419.60	
Esophagus dose (μGy)							
15<30 kg	41	0.16	1741.51	16.51	62.81	272.85	0.002
30<50 kg	96	0.09	2224.52	10.22	45.51	132.90	
50<80 kg	164	0.05	2541.01	20.54	64.14	182.53	
>=80 kg	38	0.69	4316.43	50.38	148.84	962.09	
Ovaries dose (μGy)							
15<30 kg	16	0.01	4.28	0.26	1.24	2.12	0.002
30<50 kg	44	0.00	5.97	0.04	0.21	0.47	
50<80 kg	70	0.00	6.22	0.08	0.26	1.07	
>=80 kg	10	0.02	10.34	0.34	1.27	4.60	
Remainder male dose (μGy)							
15<30 kg	25	0.05	411.06	5.08	13.92	85.42	0.160
30<50 kg	52	0.03	840.92	4.53	17.20	75.54	
50<80 kg	94	0.03	1189.69	9.65	24.28	60.88	
>=80 kg	28	0.21	1347.85	14.44	39.85	270.96	
Remainder female dose (μGy)							
15<30 kg	16	0.53	180.96	10.87	52.31	89.79	0.006
30<50 kg	44	0.06	481.63	3.37	17.09	37.61	
50<80 kg	70	0.02	751.47	9.03	30.85	129.58	
>=80 kg	10	2.23	960.76	31.89	118.42	427.46	

min. minimal value, *max.* maximal value, *N* number of participants

The EDs for different arrhythmia substrates are also shown in Table 4. There were statistically significant differences ($p < 0.001$) between different substrates. The difference was found between AVNRT and left-sided AP ($p = 0.009$), AVNRT and ventricular substrates ($p < 0.001$), right-sided and left-sided AP ($p = 0.027$), right-sided AP and ventricular substrates ($p < 0.001$), left-sided AP and ventricular substrates ($p = 0.001$), and AT and ventricular substrates ($p = 0.027$).

Reasons for the Use of X-rays

Table 5 shows the reasons for the use of X-rays and the median values of DAP and ED (with minimum, maximum, and percentiles) for all reasons. The most frequent reason

was transseptal puncture. The highest median DAP value of 43.07 (IQR 12.0–169.0) cGycm² and an ED of 173.49 (IQR 74.24–526.93) μSv were in the group of those irradiated for angiography.

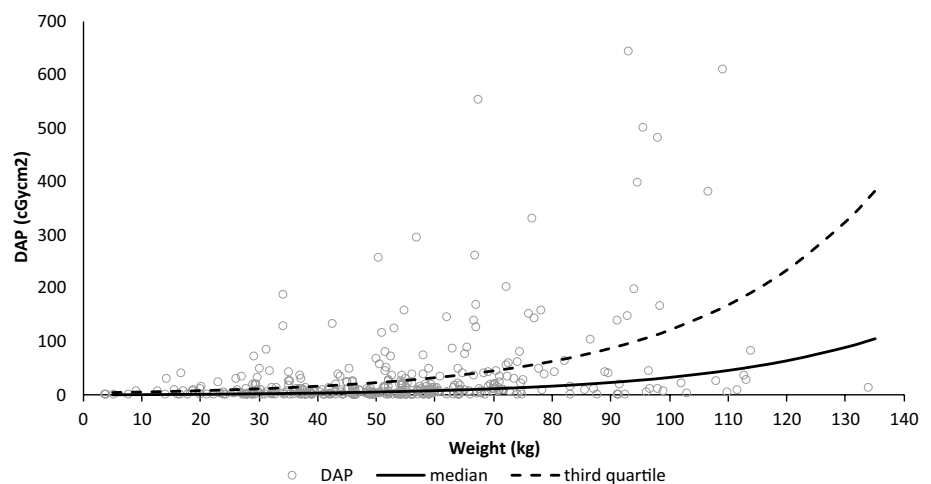
Statistically significant differences ($p < 0.001$) were found between the DAP values and EDs for the same reasons. The difference was significant between transseptal puncture and positioning of the wire, introducer or catheter ($p < 0.001$ for DAP and ED), between transseptal puncture and angiography ($p = 0.003$ for DAP and $p = 0.001$ for ED), positioning of the wire, introducer or catheter and mapping ($p < 0.001$ for DAP and ED), positioning of the wire, introducer or catheter and angiography ($p < 0.001$ for DAP and ED), and between mapping and angiography ($p = 0.018$ for DAP and $p = 0.002$ for ED).

Table 2 Dunn post hoc test for dose area product, effective dose, and organ doses between mass groups

Dunn post hoc test	<i>P</i> values					
Weight groups/Variables	15–< 30 kg vs. 30–< 50 kg	15–< 30 kg vs. 50–< 80 kg	15–< 30 kg vs. > = 80 kg	30–< 50 kg vs. 50–< 80 kg	30–< 50 kg vs. > = 80 kg	50–< 80 kg vs. > = 80 kg
Dose area product	0.722	0.002	0.000	0.000	0.000	0.007
Effective dose	0.267	0.554	0.056	0.029	0.002	0.047
Heart dose	0.171	0.791	0.053	0.024	0.000	0.018
Stomach dose	0.168	0.807	0.103	0.120	0.003	0.028
Colon dose	0.067	0.270	0.176	0.339	0.003	0.008
Liver dose	0.710	0.313	0.022	0.069	0.002	0.038
Lungs dose	0.242	0.682	0.052	0.038	0.002	0.041
Urinary bladder dose	0.009	0.000	0.724	0.644	0.011	0.000
Thyroid dose	0.048	0.260	0.251	0.244	0.004	0.015
Active Bone marrow dose	0.873	0.006	0.000	0.000	0.000	0.043
Breasts dose	0.444	0.657	0.116	0.136	0.056	0.168
Esophagus dose	0.149	0.525	0.038	0.265	0.000	0.000
Ovaries dose	0.001	0.019	0.792	0.144	0.010	0.050
Remainder male dose	0.579	0.328	0.057	0.695	0.068	0.084
Remainder female dose	0.017	0.833	0.171	0.015	0.005	0.089

Numbers in bold— $p < 0.05$

Fig. 1 Dependence of dose area product (DAP) value on patient's mass. Typical value (median) is presented with the solid line and third quartile with the dashed line



Discussion

This study showed that DAP and ED are higher in children with higher mass compared to lower mass groups. This can be explained by a larger dose of X-rays needed for an image of acceptable quality in a larger child and a larger area used to show part of the thorax that is of interest. We did not expect this result, since, in smaller children, there should be more troubleshooting, tough punctures, problems with 3D mapping system, mapping problems, and so on, which would push toward the use of X-rays, which did not happen, at least not measurable in DAP and ED.

We have to keep in mind that many procedures that were done without the use of X-rays were excluded from this study, so the overall doses of radiation for children in EP are even lower. The numbers in this study are only applicable to the cases in which the X-rays are used. Patient exposure can even be lower in centers that use intracardiac echocardiography for every case [19], but the ratio of radiation doses between mass groups should remain the same. In our centers, we do not do it because of large sheets for the intracardiac echo probe, which can be a special problem in small children, and because of the cost of such practice. This also explains why transseptal puncture is the most frequent

Table 3 Dose area product and effective dose in relation to indication

	Indication	Valid <i>N</i>	Min	Max	Perc. 25	Median	Perc. 75	<i>P</i>
Dose area product (cGycm ²)	SVT	148	0.01	609.00	2.00	7.00	27.00	< 0.001
	Preexcitation	74	0.01	294.37	1.00	4.00	12.76	
	WPW	67	0.01	642.49	4.00	11.50	38.15	
	Palpitation	30	0.04	553.00	4.00	10.00	22.00	
	PVC/VT	28	0.97	500.00	12.9	27.6	75.00	
Effective dose (μSv)	SVT	148	0.03	1700.90	11.12	32.14	108.85	< 0.001
	Preexcitation	74	0.04	1082.64	3.74	18.09	56.70	
	WPW	67	0.07	1767.88	13.07	39.13	146.67	
	Palpitation	30	0.17	1713.97	14.61	31.73	68.20	
	PVC/VT	28	3.54	1339.79	53.31	100.73	258.10	

min. minimal value, *max.* maximal value, *N* number of participants, *perc.* percentile, *PVC* premature ventricular contractions, *SVT* supraventricular tachycardia, *WPW* Wolff–Parkinson–White syndrome, *VT* ventricular tachycardia

Table 4 Dose area product and effective dose in relation to arrhythmia substrate

	Arrhythmia substrate	Valid <i>N</i>	Min	Max	Perc. 25	Median	Perc. 75	<i>P</i>
Dose area product (cGycm ²)	AVNRT	30	0.01	553.00	1.00	3.68	5.88	< 0.001
	Right-sided AP	70	0.01	256.03	1.00	4.80	19.32	
	Left-sided AP	200	0.01	642.49	3.00	10.00	27.50	
	AT	19	0.10	330.00	4.00	11.00	41.00	
	ventricular	27	0.97	500.00	13.81	29.62	76.00	
Effective dose (μSv)	AVNRT	30	0.03	1713.97	9.01	14.15	23.98	< 0.001
	Right-sided AP	70	0.04	1063.31	4.49	19.58	73.50	
	Left-sided AP	200	0.05	1767.88	14.36	37.60	127.02	
	AT	19	0.28	901.14	10.94	56.70	117.23	
	ventricular	27	3.54	1339.79	60.78	103.15	266.99	

AP accessory pathway, *AT* atrial tachycardia, *AVNRT* atrioventricular nodal reentry tachycardia, *cGy* Centigray, *min.* minimal value, *max.* maximal value, *N* number of participants, *perc.* percentile

Table 5 Dose area product and effective dose in relation to fluoro use reason

	Fluoro use reason	Valid <i>N</i>	Min	Max	Perc. 25	Median	Perc. 75	<i>P</i>
Dose area product (cGycm ²)	TS	219	0.01	642.49	3.00	10.00	31.59	< 0.001
	posit	73	0.01	126.00	1.00	2.00	9.00	
	map	40	0.10	256.03	2.50	15.83	34.88	
	angio	15	0.01	553.00	12.00	43.07	169.00	
Effective dose (μSv)	TS	219	0.05	1767.88	14.92	42.97	135.10	< 0.001
	posit	73	0.04	392.86	4.52	12.86	28.58	
	map	40	0.28	1063.31	11.37	62.32	107.66	
	angio	15	0.03	1713.97	74.24	173.49	526.93	

min. minimal value, *max.* maximal value, *N* number of participants, *perc.* percentile, *posit* positioning of the wire, introducer or catheter, *map* mapping, *TS* transseptal puncture

reason for the use of X-rays and measured radiation doses for the left accessory pathways in our study.

Of all the reasons for X-ray use, angiography had the highest DAP and ED which is most likely because of the need for higher image quality when doing angiography, which demands higher doses of X-rays.

Compared to the diagnostic reference levels for pediatric interventional cardiology that were developed for Latin America [23], our results for electrophysiological procedures show significantly lower levels. Ubeda et al. [23] showed DAP median values for different procedures put together in mass groups: 15–30 kg 3.3 Gy·cm², 30–50 kg 6.2 Gy·cm²,

and 50–80 kg $23.6 \text{ Gy}\cdot\text{cm}^2$. These values are 10 times and more higher than in this study (0.04 to $0.26 \text{ Gy}\cdot\text{cm}^2$).

Buytaert et al. [21] also showed DAP values for pediatric interventional cardiology. In that study, the median values of DAP for diagnostic and interventional cardiac procedures were 200.31 and 189.29 cGycm^2 , respectively [21], which is quite above our values. They [21] also showed that the ED for diagnostic cardiac procedures is 3.34 mSv and for interventional 1.6 mSv, which is significantly more than the values for pediatric electrophysiology.

The values in our study were lower than what Bacher et al. [24] showed also for pediatric interventional cardiology. In this article, median DAP values ranged from 272 to 548 cGycm^2 and median EDs ranged from 4.6 to 6.0 mSv [24].

Riche et al. [25] in their study investigated radiation dose when using low fluoroscopy frame rate in pediatric electrophysiology, but without 3D mapping systems. The DAP median value for 168 patients was 246 cGycm^2 [25], which is, although quite low for X-ray-based electrophysiology, significantly higher than our results. In the same article [25], ED was calculated for 14 patients and the median was 0.19 mSv, also higher than our results. A similar study from Gellis et al. [26] with 42 patients younger than 21 years, using only X-rays, showed a median DAP of 343.2 cGycm^2 which is higher than what Riche et al. [25] described. An even higher value of DAP was noted by Patel et al. [27]. In that article, the median DAP was 1741.4 cGycm^2 after the implementation of the new protocol for the reduction of radiation, but all procedures were performed without the use of 3D mapping system [27]. They also showed a median ED of 4.6 mSv [27], which is a substantially higher value than presented in our article. These relations are another proof of the importance of 3D mapping systems in pediatric electrophysiology.

Pass et al. [28] showed a reduction in radiation dose after the introduction of CARTO system in 75 pediatric patients. The DAP mean decreased from 584.3 to 92.2 cGycm^2 with the use of the 3D mapping system. They did not make a difference based on patients' age or mass. In a study from Jiang et al. [15], DAP mean value for AVNRT ablation was 51.4 cGycm^2 . The difference in the last two mentioned studies is that they used the Carto system for all the procedures.

We have not found any studies on EDs in larger number of pediatric electrophysiology procedures while using the 3D mapping system.

Compared to X-ray use in adult electrophysiology procedures without 3D mapping systems [29, 30], the doses in our study are quite lower. Even for diagnostic procedures, they showed the median ED was 3.2 mSv, and AVNRT, atrial tachycardia, and atrioventricular reentry tachycardia ablation 4.4 mSv [29]. In the NO-PARTY trial [31], a reduction of the DAP and ED with the use of a 3D mapping system was

noted. In this trial [31], they included adults in whose cases no X-rays were used so the values cannot be directly compared to ours, but the importance of 3D mapping is visible: DAP median went from 2036 to 278 cGycm^2 with the use of 3D mapping and ED from 8.87 to 0 (IQR 0–0.08) mSv.

To get a better sense of the order of magnitude of ED values, one can compare it to different general radiographic images in children that are shown by Earl et al. [32]. In that article, the estimated ED for anterior and posterior chest X-ray, in different weight ranges, was 0.007–0.033 mSv, or chest lateral was 0.007–0.047 mSv, which is less than pediatric electrophysiology, but not as much as we expected. Already the anterior posterior X-ray of the abdomen has estimated EDs (0.03–0.286 mSv) comparable to or higher than pediatric electrophysiology.

Limitations

This was a cross-sectional study with its downsides. We did not develop new ED CFs but used previously calculated ones as they are applicable for this study since during pediatric electrophysiology procedures, the same area of the thorax is irradiated as in interventional cardiology for which these factors were developed. Also, estimated CFs from other papers are similar in relation to age, so our comparison of age groups and other variables from our data should also be applicable [33]. We used DAP as the main dose parameter, but it alone cannot fully describe this complex matter. Nevertheless, this parameter is easily available and most often used in the description of the radiation dose in fluoroscopy. This makes it most suitable for the comparison of different institutions. Another shortcoming of our work is the lack of information on the exact X-ray tube position, that is, how much radiation the patient received during specific projection, which would enable us to determine the organ doses and effective doses more precisely.

Conclusion

Pediatric electrophysiology can be done with very low doses of X-rays when using 3D mapping systems compared to X-ray-based electrophysiology, or when compared to pediatric interventional cardiology or adult electrophysiology. There is a statistical difference in doses based on mass groups, with higher doses in larger groups. Based on this study, there are no arguments for postponing electrophysiological procedures to older age due to the X-ray exposure.

Author Contributions NK analyzed the data, performed literature search, and wrote the manuscript; LK gave the idea for the study and

helped in writing the manuscript; IK oversaw the project and helped in writing the manuscript; DDB oversaw the project and helped in writing the manuscript; MM performed the statistical analysis and analyzed the data; MR and FK contributed to the data collection and literature search; and VR oversaw the project, helped in writing the manuscript, and analyzed the data. All authors read and approved the final manuscript.

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Declarations

Competing interest The authors have no competing interests to declare that are relevant to the content of this article.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. The study has been reviewed and approved by ethics committees in Croatia (Ethics committee, University hospital center Sestre milosrdnice—registry number 251-29-11-21-05) and in Hungary (Health Science Council, Scientific and Research Ethics Committee, ETT TUKEB—registry number BMEÜ/3824-3/2022/EKU).

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